

Informe de experimento: single_call_structured

Generado el 2026-09-18 21:59 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio single_call_structured (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.6-sol
- Métodos: single_call:structured
- Casos: case14
- Requests: generadas, N=10 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=3, parameterized=3, multistep=2, ambiguous=2
- System prompt: hash 23e5d9b407f9 (10 filas)
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-18 04:11
- Costo total: 0.1161 USD (10 items; costo medio 0.01161 USD/item)
- Items totales: 10; ok=10; con error=0; tokens totales=47832
- Solved global: 10.0% (1/10)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método single_call:structured (system_prompt_hash real de esta fila: 23e5d9b407f9)

System prompt (reconstrucción con el código actual, no necesariamente histórica) —1879 caracteres; strategy=structured; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
```

3. Violations, if any
4. Suggested next step

Single-call mode

You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns, and do not fabricate numbers.

User prompt (primer item) —3821 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)
```

Branches

```
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)
```

Available tools

- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Answer by emitting tool calls (function calling) in THIS single response:

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a non-converged result, say so.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), `V_MAE` (p.u., solo sobre peticiones con formulación exacta —toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y `F_MAE` (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a

extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_automously_rate (no la columna Solved de arriba, que puede solaparse con Escalated –una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100 %: dónde termina cada petición –resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case N	Form.	V_M	KEM	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Escalated	Wrong (silent)	\$
single_call:structured	case1410	10.0 (1/10)	0	2.60	10.0 (1/10)	100.0 (10/10)	2.28e-02	100.0 (10/10)	–	1.0	4783–	0.0 (0/10)	90.0 (9/10)	0.1161	

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
single_call:structured	33.3 (1/3)	0.0 (0/3)	0.0 (0/2)	0.0 (0/2)

Solved por dificultad (%, items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
single_call:structured	33.3 (1/3)	0.0 (0/3)	0.0 (0/2)	0.0 (0/2)

Tipos de error de formulación (conteo de items)

Method	missed_step	ok
single_call:structured	9	1

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de traces/ cuando existe.

Método single_call:structured

Ejemplo: resuelto (solved=True)

```
request (case14-plain-000-s0, case14, difficulty=plain, seed=1813382119): Load case14 and run the power flow, then report the bus with the
↳ lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: The case14 network loaded successfully, but no power-flow result was returned, so the lowest-voltage bus cannot be identified.

**Key numbers**
- Total load: **252.6562386075145 MW**
```

```
- Total generation: **Not available** (only generation capacity of **772.4000000000001 MW** was reported)
- Total los ...[223 chars más]
--- Respuesta final ---
The case14 network loaded successfully, but no power-flow result was returned, so the lowest-voltage bus cannot be identified.
```

```
**Key numbers**
- Total load: **252.6562386075145 MW**
- Total generation: **Not available** (only generation capacity of **772.4000000000001 MW** was reported)
- Total losses: **Not available**
- Voltage range: **Not available**
```

```
**Violations:** Cannot be assessed without a converged power flow.
```

```
**Suggested next step:** Run `run_powerflow()` and inspect the resulting bus voltage magnitudes.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4771;
↳ cost_usd=0.0117; wall=12.9 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: Case14 was loaded successfully, but the requested bus 10-11 line disconnection was not performed.
```

```
- **Total load:** 253.84578240591338 MW
- **Total generation capacity:** 772.4000000000001 MW
- **Total generation, losses, voltage range:** Not available because no power flow was run
- **Violat ...[121 chars más]
--- Respuesta final ---
Case14 was loaded successfully, but the requested bus 10-11 line disconnection was not performed.
```

```
- **Total load:** 253.84578240591338 MW
- **Total generation capacity:** 772.4000000000001 MW
- **Total generation, losses, voltage range:** Not available because no power flow was run
- **Violations:** Not assessed
```

```
**Suggested next step:** Disconnect the branch between buses 10 and 11 and re-run the power flow.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=2.05e-03; F_MAE=1.39; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=4725; cost_usd=0.0113; wall=7.2 s
```

Ejemplo: no resuelto

```
request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 261.9028323717408,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. **Summary:** IEEE case14 was loaded successfully, but the requested 6-13 line disconnection was not executed.
2. **Key numbers:** 14 buses, 5 generators, 20 branches, 11 loads; total load **261.902832 MW**; generation capacity **772.4 MW**.
↳ Power-flow generation, losses, and voltage range were no ...[196 chars más]
--- Respuesta final ---
1. **Summary:** IEEE case14 was loaded successfully, but the requested 6-13 line disconnection was not executed.
2. **Key numbers:** 14 buses, 5 generators, 20 branches, 11 loads; total load **261.902832 MW**; generation capacity **772.4 MW**.
↳ Power-flow generation, losses, and voltage range were not reported.
3. **Violations:** Not assessed because no post-disconnection power flow was run.
4. **Suggested next step:** Disconnect the branch between buses 6 and 13 and recompute the power flow.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=8.99e-03; F_MAE=3.80; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=4755; cost_usd=0.0114; wall=8.3 s
```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 1 filas método×caso con solver: single_call:structured/case14 (3.35e-03).
- El gate no verificó ningún resultado de flujo (gate_checked=0 en todos los items).
- Ningún item terminó con error.
- Solved de single_call:structured: 10.0% (1/10).

- Formulación exacta agregada sobre métodos con tools: 10.0 % (1/10).
- Formulación por dificultad: mejor `plain` 33.3 % (1/3), peor `parameterized` 0.0 % (0/3).
- Errores de formulación más frecuentes: `missed_step=9`.
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 0.1161 USD; el método más caro fue `single_call:structured` con 0.1161 USD (100 %).
- Métodos con tools: 1.00 tool calls medias por item (máximo 1).

6. Archivos

Rutas absolutas bajo `single_call_structured.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard.json
↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard.csv
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard.md
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard_per_case.json
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/scoreboard_per_case.json
traces (10 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/traces
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/single_call_structured/traces
```